NFDI4Culture Integration Stories: Bridging Gaps Between Isolated Research Resources

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Cultural heritage (CH) data are rich resources helping researchers to better understand CH networks and reveal hidden connections within interdisciplinary and multimodal datasets. They range from musical works and sheet music, images and paintings, historical landmarks and buildings, performances and events to 3D models and textual sources. Although authority files and controlled vocabularies, such as the German Integrated Authority File (GND), VIAF , Wikidata , and Iconclass are extensively used, most of the CH data lacks common indexing and is not readily accessible for querying or federation. This makes navigating various datasets challenging.

NFDI4Culture Knowledge Graph

NFDI4Culture is the consortium within the German National Research Data Infrastructure programme (NFDI) that is establishing an infrastructure for material and immaterial CH data. Its primary goal is to ensure the findability and interoperability of distributed and heterogeneous

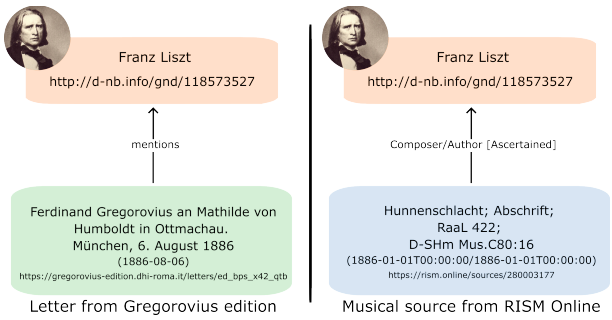
research data across five subject areas: architecture, art history, performing arts, musicology, and media studies. To achieve this, the NFDI4Culture Knowledge Graph (NFDI4Culture-KG) is being developed. Metadata about diverse and isolated research data are aggregated, thereby increasing the findability, interoperability and reusability of CH data through a KG-based research data index (Tietz et al. 2023, Bruns et al. 2024b). The NFDI4Culture-KG currently contains metadata about over 8 million resources.[1] It is accessible via a SPARQL endpoint as well as, perspectively, a searchable graphical user interface.

The ongoing expansion of the NFDI4Culture-KG can be showcased by the integration of metadata about research data from the digital edition *Ferdinand Gregorovius: Poesie und Wissenschaft. Gesammelte deutsche und italienische Briefe* of the German Historical Institute Rome (DHI Rom) and their interconnections to musicological data from *RISM Online* , a service by RISM Digital .

Source Data

RISM Online is an access point to over 1,563,000 musical sources of the Répertoire International des Sources Musicales . Users can access the data via various filter options and an incipit search. The portal offers an API to request data as JSON-LD via HTTP content negotiation in accordance with schema.org .[2] The individual sources are enriched with external identifiers, such as VIAF and GND IDs.

The Gregorovius edition contains 1,093 annotated pieces of correspondence from the historian Ferdinand Gregorovius (1821-1881), whose heritage from the 19th century testifies a rich engagement with intellectual-historical movements and musicians of his time. The letters are edited in TEI-XML and enriched with GND and GeoNames IDs. In addition to a graphical user interface, resources are made available via various APIs.[3] Besides BEACON and CMIF (Correspondence Metadata Interchange Format) the edition offers data requests for individual resource types such as letters, persons, places and works in JSON via a RESTful API.

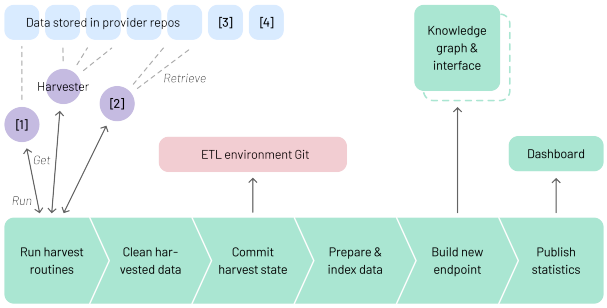


Example reference of F. Liszt in Gregorovius edition and RISM online. Creator: L. Söhn, T. Tietz.

Through the use of external identifiers and based on personal, temporal and spatial references, the data from the Gregorovius edition and RISM Online are related in terms of content (Figure 1).[4] However, prior to the integration into the NFDI4Culture-KG, it was not possible to query or search these via a common access point.

Integration Story

In order to create an overarching index for different datasets, the NFDI4Culture Ontology (CTO) is applied as a lightweight approach in the NFDI4Culture-KG (Tietz et al. 2025). It is designed to describe the specifics of individual resources, e.g. music incipits, as well as elements connecting them, such as people, times, places, organisations and concepts, and to link back to the source data. IDs of authority data and controlled vocabularies in the sources themselves play a decisive role in making cross-connections discoverable, even beyond the specific identifiers in the source data.



Data Integration:[1]RISM-Harvester, [2]Greg.-Jupyter-Notebook, [3]RISM-Online, [4]Greg. edition. Creator: J. J. Steller.

Data integration is managed through an Extract-Transform-Load (ETL) environment (Figure 2), the so-called Culture Knowledge Graph Kit-

chen , to harvest data from various endpoints and transform it CTO-compatible (Bruns et al. 2024a, Schrade / Sack 2024, Steller et al. 2024). The ETL-environment consists of six modules that can be used in combination or independently.

1: **Executing harvesting routines** based on RDF-action files with step definitions to harvest data with external tools.

2: **Cleaning the collected data** by harmonisation with the associated action file, triples representing the harvest state are added or deleted.

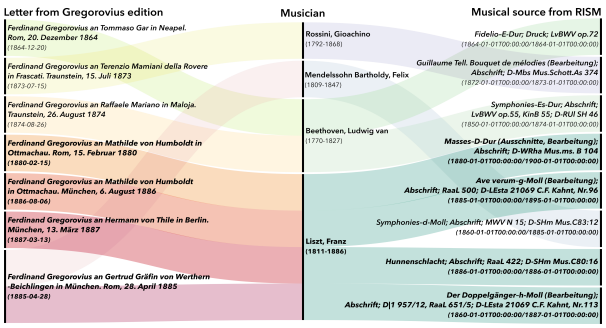
3: **Committing the harvesting status** by pushing changes made during a harvesting run to the pipeline repository.

4: **Preparing and indexing data** by updating or creating data directories and search indexes.

5: **Creating a new endpoint** with zero downtime by deploying a new triple store container when the new endpoint is operational.

6: **Publishing statistics** about the integrated data feeds via a dashboard with visualisations based on provided SPARQL-queries.

This setup allows different tools for harvesting and transformation to be integrated into the first module. While a comprehensive harvester, the Hydra Scraper , is being developed for the integration of research data provided according to standards, such as the Lightweight Information Describing Objects (LIDO) or the Culture Graph Interchange Format (CGIF), other tools can be used according to the requirements of data providers, or data dumps can be requested directly.



Example interconnections of F. Liszt et al. in Greg. edition and RISM Online (SPARQL Query). Creator: L. Söhn.

A custom harvester for RISM Online was developed that fetches the data in JSON-LD and transforms it CTO-compatible.[5] In contrast, a simple transformation workflow using a Jupyter notebook was applied to integrate metadata from the JSON-API of the Gregorovius edition.[6]

It is now possible to find interconnections between musical sources from RISM Online and letters from the Gregorovius edition e.g. via related persons in the NFDI4Culture-KG (Figure 3). This enables researchers to carry out analyses via the graph itself or to continue working with the more detailed source data from the providers.

Notes

1. The number given corresponds to the status on 9 May 2025. Current statistics on resources and statements can be viewed on the public dashboard of the NFDI4Culture-KG at .
2. The documentation of the RISM Online API can be found at .
3. The Gregorovius Correspondence Edition API is documented at .
4. All figures in this document are licensed under a CC-BY 4.0 licence.
5. The RISM Online Harvester is available under .
6. The jupyter notebook to harvest data from the Gregorovius Correspondence Edition API can be found under .

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